Lecture 7: Transportation Problems

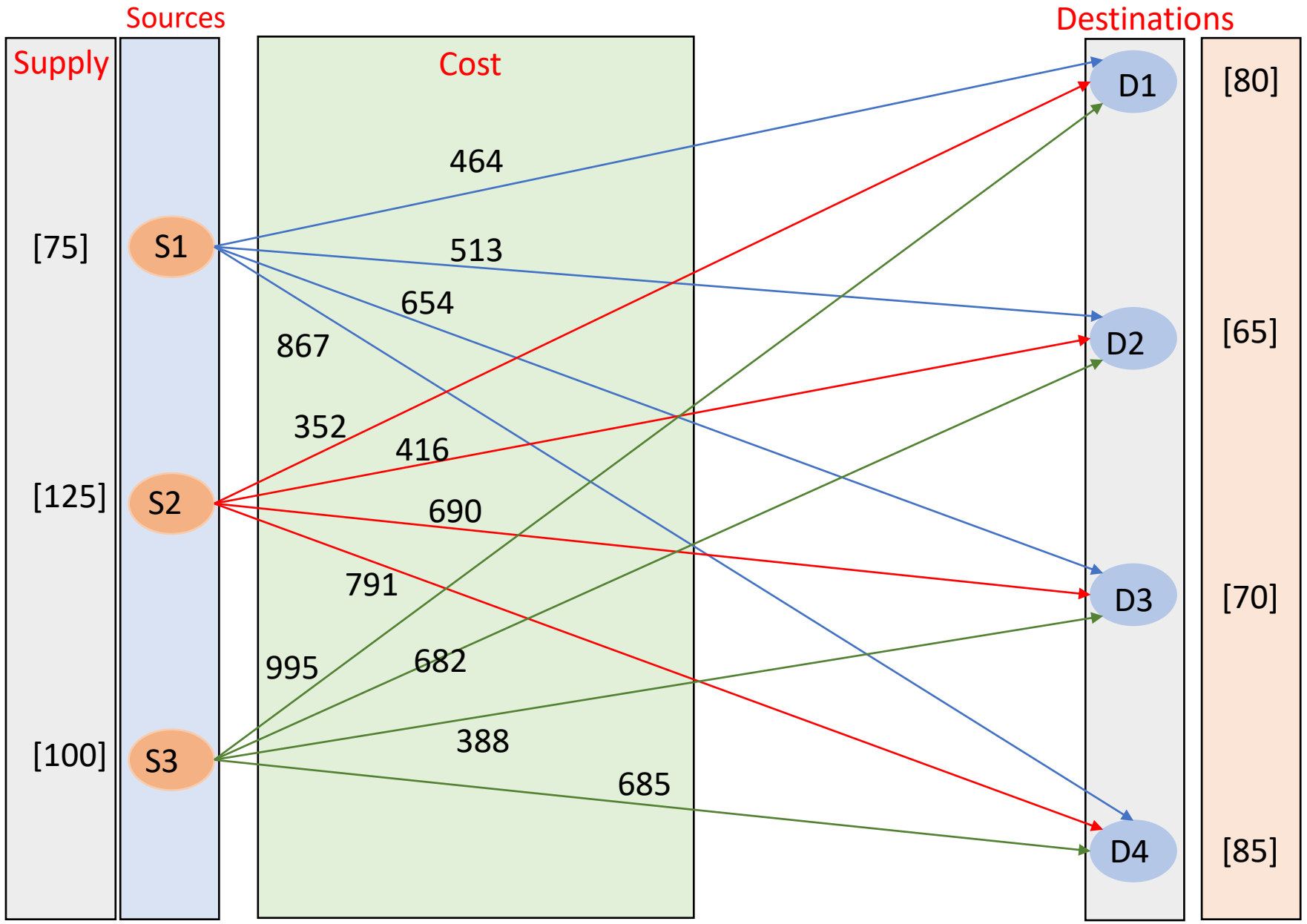
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Course of Introduction to Optimization

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Transportation Problems



Terminology for the transportation problem

Units of a commodity

m sources

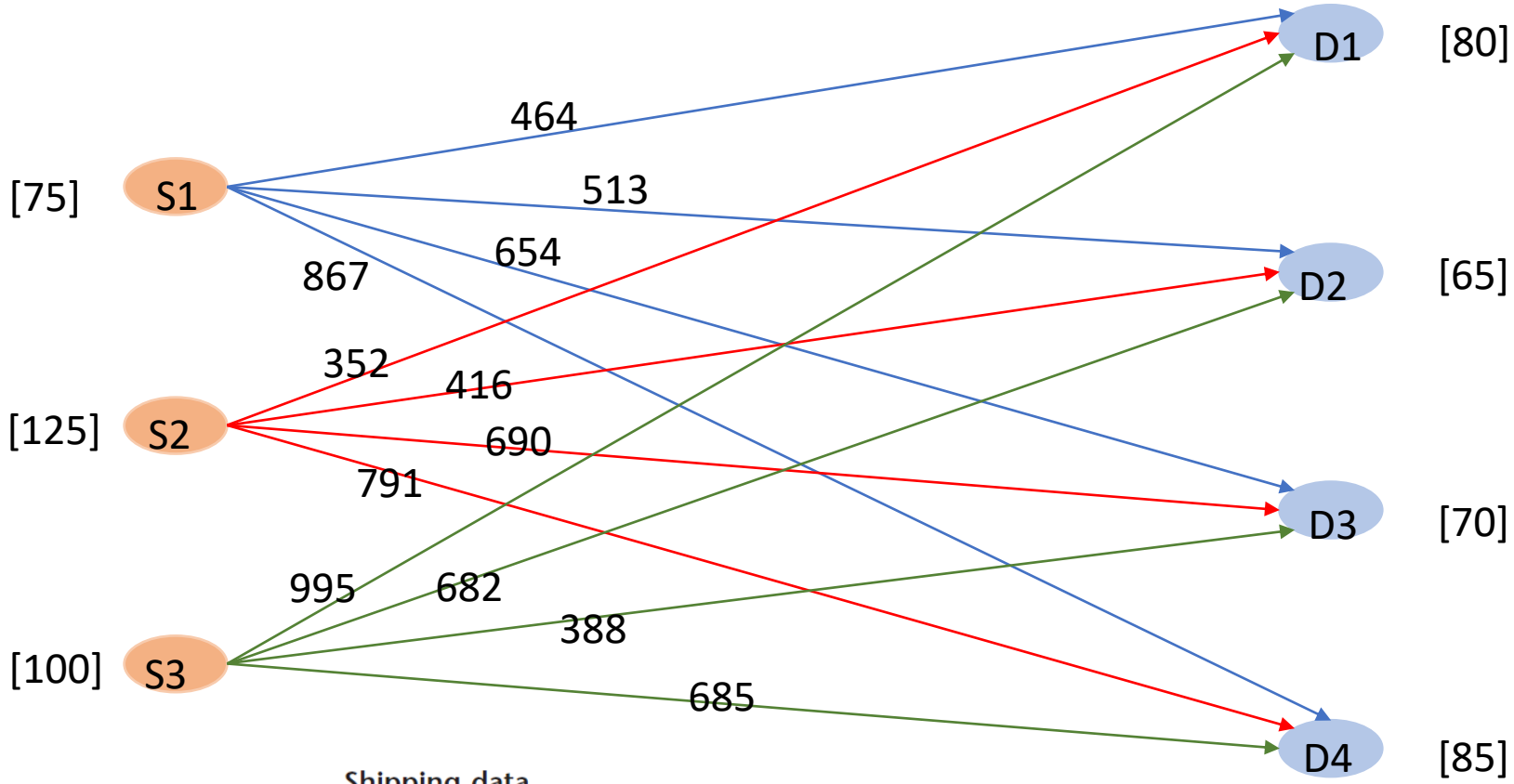
n destinations

Supply *si* from source *i*

Demand *dj* at destination *j*

Cost *cij* per unit distributed from source *i* to destination *j*

Transportation Problems



Shipping data

	Shipping Cost ($) per Truckload				
	Warehouse (Destinations)				Supply
	1	2	3	4	Output
Sources 1	464	513	654	867	75
2	352	416	690	791	125
3	995	682	388	685	100
Allocation	80	65	70	85	

Transportation Problems

Shipping data

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	Warehouse (Destinations)				Supply
	1	2	3	4	Output
Sources 1	464	513	654	867	75
2	352	416	690	791	125
3	995	682	388	685	100
Allocation	80	65	70	85	

Thus, the objective is to choose the values of these 12 decision variables (the x_{ij}) so as to

$$\text{Minimize} \quad Z = 464x_{11} + 513x_{12} + 654x_{13} + 867x_{14} + 352x_{21} + 416x_{22} + 690x_{23} + 791x_{24} + 995x_{31} + 682x_{32} + 388x_{33} + 685x_{34},$$

subject to the constraints

$$\begin{aligned}
x_{11} + x_{12} + x_{13} + x_{14} &= 75 \\
x_{21} + x_{22} + x_{23} + x_{24} &= 125 \\
x_{31} + x_{32} + x_{33} + x_{34} &= 100 \\
x_{11} + x_{21} + x_{31} &= 80 \\
x_{12} + x_{22} + x_{32} &= 65 \\
x_{13} + x_{23} + x_{33} &= 70 \\
x_{14} + x_{24} + x_{34} &= 85
\end{aligned}$$

and

$$x_{ij} \geq 0 \qquad (i = 1, 2, 3; j = 1, 2, 3, 4).$$

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\end{aligned}$$

and

$$x_{ij} \geq 0 \qquad (i = 1, 2, 3;\ j = 1, 2, 3, 4).$$

Constraint coefficients

Coefficient of:

	x_{11}	x_{12}	x_{13}	x_{14}	x_{21}	x_{22}	x_{23}	x_{24}	x_{31}	x_{32}	x_{33}	x_{34}	
A =	1	1	1	1									Supply constraints
					1	1	1	1					
									1	1	1	1	
	1				1				1				Demand constraints
		1				1				1			
			1				1				1		
				1				1				1	

Transportation Problem Model

To describe the general model for the transportation problem, we need to use terms that are considerably less specific than those for the components of the prototype example.

In particular, the general transportation problem is concerned (literally or figuratively) with distributing *any* commodity from *any* group of supply centers, called **sources**, to *any* group of receiving centers, called **destinations,** in such a way as to minimize the total distribution cost.

As indicated by the fourth and fifth rows of the table, each source has a certain **supply** of units to distribute to the destinations, and each destination has a certain **demand** for units to be received from the sources.

The model for a transportation problem makes the following assumption about these supplies and demands:

Terminology for the transportation problem

Units of a commodity

m sources

n destinations

Supply *si* from source *i*

Demand *dj* at destination *j*

Cost *cij* per unit distributed from source *i* to destination *j*

Transportation Problem Model

The model for a transportation problem makes the following assumption about these supplies and demands:

The requirements assumption: Each source has a fixed *supply* of units, where this entire supply must be distributed to the destinations. (We let *si* denote the number of units being supplied by source *i*, for $i = 1, 2, \ldots, m$.) Similarly, each destination has a fixed *demand* for units, where this entire demand must be received from the sources. (We let *dj* denote the number of units being received by destination *j*, for $j = 1, 2, \ldots, n$.)

The feasible solutions property: A transportation problem will have feasible solutions if and only if

$$\sum_{i=1}^{m} s_i = \sum_{j=1}^{n} d_j.$$

In some real problems, the supplies actually represent *maximum* amounts (rather than fixed amounts) to be distributed. Similarly, in other cases, the demands represent maximum amounts (rather than fixed amounts) to be received. Such problems do not quite fit the model for a transportation problem because they violate the *requirements assumption*. However, it is possible to *reformulate* the problem so that they then fit this model by introducing a *dummy destination* or a *dummy source* to take up the slack between the actual amounts and maximum amounts being distributed.

Transportation Problem Model

The cost assumption: The cost of distributing units from any particular source to any particular destination is *directly proportional* to the number of units distributed.

Therefore, this cost is just the *unit cost* of distribution *times* the *number of units distributed.* (We let c_{ij} denote this unit cost for source i and destination j.)

The only data needed for a transportation problem model are the supplies, demands, and unit costs. These are the *parameters of the model.* All these parameters can be summarized conveniently in a single *parameter table* as shown in Table

Parameter table for the transportation problem

		Cost per Unit Distributed				
		Destination				
		1	**2**	⋯	***n***	**Supply**
Source	1	c_{11}	c_{12}	⋯	c_{1n}	s_1
	2	c_{21}	c_{22}	⋯	c_{2n}	s_2
	⋮	⋯⋯	⋯⋯	⋯⋯	⋯⋯	⋮
	m	c_{m1}	c_{m2}	⋯	c_{mn}	s_m
Demand		d_1	d_2	⋯	d_n	

Transportation Problem Model

Parameter table for the transportation problem

		Cost per Unit Distributed				
		Destination				
		1	2	⋯	n	Supply
Source	1	c_{11}	c_{12}	⋯	c_{1n}	s_1
	2	c_{21}	c_{22}	⋯	c_{2n}	s_2
	⋮	⋮	⋮	⋮	⋮	⋮
	m	c_{m1}	c_{m2}	⋯	c_{mn}	s_m
Demand		d_1	d_2	⋯	d_n	

$$\text{Minimize} \quad Z = \sum_{i=1}^{m} \sum_{j=1}^{n} c_{ij} x_{ij},$$

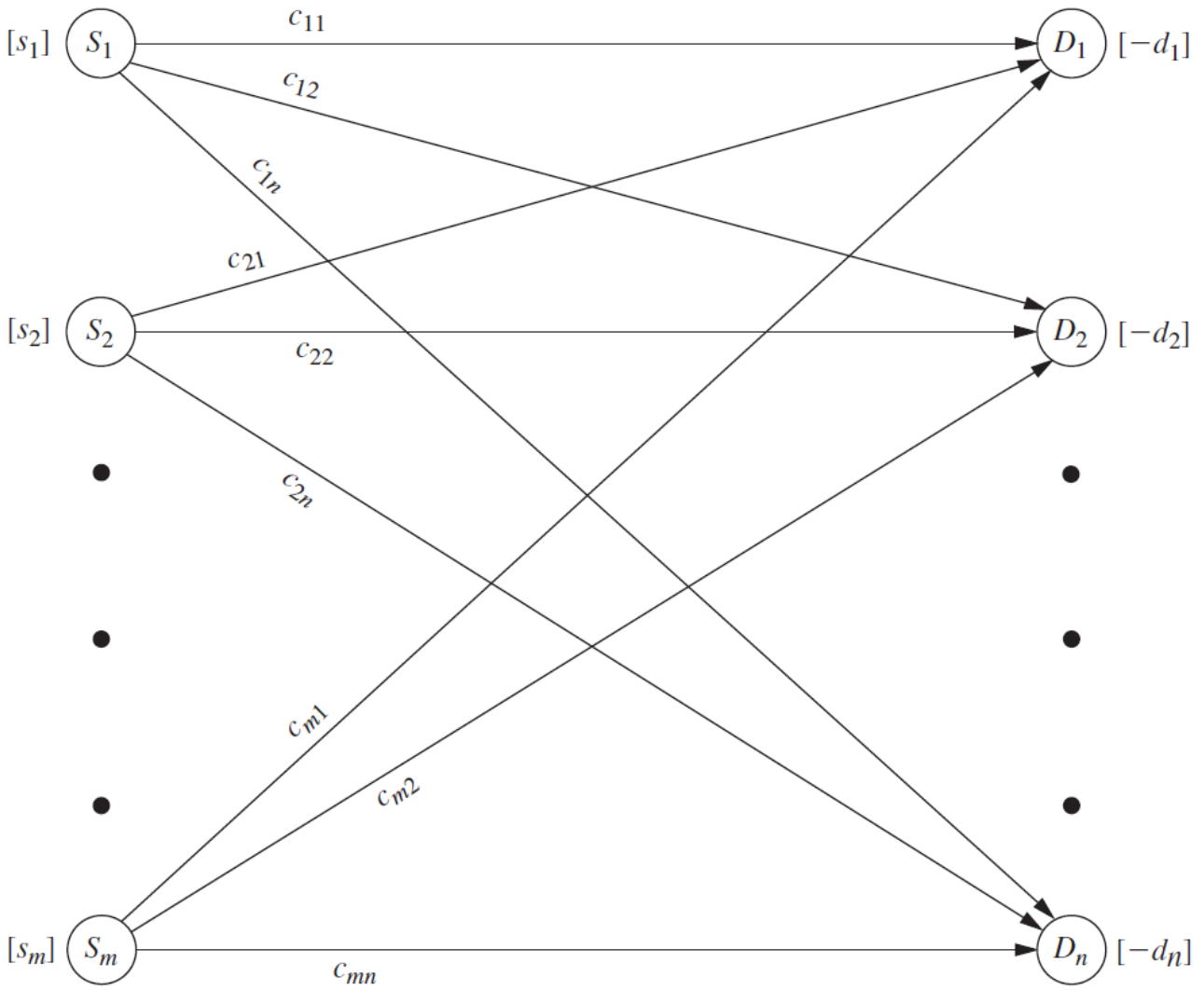
subject to

$$\sum_{j=1}^{n} x_{ij} = s_i \qquad \text{for } i = 1, 2, \ldots, m,$$

$$\sum_{i=1}^{m} x_{ij} = d_j \qquad \text{for } j = 1, 2, \ldots, n,$$

and

$$x_{ij} \geq 0, \qquad \text{for all } i \text{ and } j.$$



Setting Up the Transportation Simplex Method

To highlight the streamlining achieved by the transportation simplex method, let us first review how the general (unstreamlined) simplex method would set up a transportation problem in tabular form. After constructing the table of constraint coefficients, converting the objective function to maximization form, and using the Big M method to introduce artificial variables $z_1, z_2, \ldots, z_{mn}$ into the $m + n$ respective equality constraints, typical columns of the simplex tableau would have the form shown in Table below, where all entries *not shown* in these columns are *zeros*. [The one remaining adjustment to be made before the first iteration of the simplex method is to algebraically eliminate the nonzero coefficients of the initial (artificial) basic variables in row 0.]

Original simplex tableau before simplex method is applied to transportation problem

Basic Variable	Eq.	Coefficient of:								Right side
		Z	$\cdots$	x_{ij}	$\cdots$	z_i	$\cdots$	z_{m+j}	$\cdots$	
Z	(0)	-1		c_{ij}		M		M		0
	(1)									
	$\vdots$									
z_i	(i)	0		1		1				s_i
	$\vdots$									
z_{m+j}	$(m + j)$	0		1				1		d_j
	$\vdots$									
	$(m + n)$									

u_i multiple of *original* row i that has been subtracted (directly or indirectly) from *original* row 0 by the simplex method during all iterations leading to the current simplex tableau.

v_j multiple of *original* row $m + j$ that has been subtracted (directly or indirectly) from *original* row 0 by the simplex method during all iterations leading to the current simplex tableau.

Row 0 of simplex tableau when simplex method is applied to transportation problem

Basic Variable	Eq.	Coefficient of:								Right Side
		Z	$\cdots$	x_{ij}	$\cdots$	z_i	$\cdots$	z_{m+j}	$\cdots$	
Z	(0)	-1		$c_{ij} - u_i - v_j$		$M - u_i$		$M - v_j$		$-\sum_{i=1}^{m} s_i u_i - \sum_{j=1}^{n} d_j v_j$

The Needed Information.

To lay the groundwork for simplifying this setup, recall what information is needed by the simplex method. In the initialization, an initial BF solution must be obtained, which is done artificially by introducing artificial variables as the initial basic variables and setting them equal to s_i and d_j.

The optimality test and **step 1** of an iteration (selecting an entering basic variable) require knowing the current row 0, which is obtained by subtracting a certain multiple of another row from the preceding row 0.

Step 2 (determining the leaving basic variable) must identify the basic variable that reaches zero first as the entering basic variable is increased, which is done by comparing the current coefficients of the entering basic variable and the corresponding right side.

Step 3 must determine the new BF solution, which is found by subtracting certain multiples of one row from the other rows in the current simplex tableau.

Transportation Simplex method

Format of a transportation simplex tableau

		Destination					
		1	**2**	⋯	***n***	**Supply**	u_i
Source	1	c_{11}	c_{12}	⋯	c_{1n}	s_1	
	2	c_{21}	c_{22}	⋯	c_{2n}	s_2	
	⋮	⋯	⋯	⋯	⋯	⋮	
	m	c_{m1}	c_{m2}	⋯	c_{mn}	s_m	
Demand		d_1	d_2	⋯	d_n	$Z =$	
	v_j						

Additional information to be added to each cell:

If x_{ij} is a basic variable	*If x_{ij} is a nonbasic variable*
c_{ij} (x_{ij})	c_{ij} $c_{ij} - u_i - v_j$

General Procedure for Constructing an Initial BF Solution. To begin, all source rows and destination columns of the transportation simplex tableau are initially under consideration for providing a basic variable (allocation).

1. From the rows and columns still under consideration, select the next basic variable (allocation) according to some criterion.

2. Make that allocation large enough to exactly use up the remaining supply in its row or the remaining demand in its column (whichever is smaller).

3. Eliminate that row or column (whichever had the smaller remaining supply or demand) from further consideration. (If the row and column have the same remaining supply and demand, then arbitrarily select the *row* as the one to be eliminated. The column will be used later to provide a *degenerate* basic variable, i.e., a circled allocation of zero.)

4. If only one row or only one column remains under consideration, then the procedure is completed by selecting every *remaining* variable (i.e., those variables that were neither previously selected to be basic nor eliminated from consideration by eliminating their row or column) associated with that row or column to be basic with the only feasible allocation. Otherwise, return to step 1.

Alternative Criteria for Step 1

1. *Northwest corner rule:* Begin by selecting x_{11} (that is, start in the northwest corner of the transportation simplex tableau). Thereafter, if x_{ij} was the last basic variable selected, then next select $x_{i,j+1}$ (that is, move one column to the *right*) if source *i* has any supply remaining. Otherwise, next select $x_{i+1,j}$ (that is, move one row *down*).

Initial BF solution from the Northwest Corner Rule

		Destination						
		1	**2**	**3**	**4**	**5**	**Supply**	u_i
Source	1	16 (30)	16 (20)	13	22	17	50	
	2	14	14 (0)	13 (60)	19	15	60	
	3	19	19	20 (10)	23 (30)	M (10)	50	
	4(*D*)	M	0	M	0	0 (50)	50	
Demand		30	20	70	30	60	$Z = 2{,}470 + 10M$	
	v_j							

2. *Vogel's approximation method:* For each row and column remaining under consideration, calculate its **difference,** which is defined as *the arithmetic difference between the smallest and next-to-the-smallest unit cost* c_{ij} *still remaining in that row or column.* (If two unit costs tie for being the smallest remaining in a row or column, then the difference is 0.) In that row or column having the largest difference, select the variable having the smallest remaining unit cost. (Ties for the largest difference, or for the smallest remaining unit cost, may be broken arbitrarily.)

Transportation Simplex method

Initial BF solution from Vogel's approximation method

		Destination						
		1	**2**	**3**	**4**	**5**	**Supply**	**Row Difference**
Source	1	16	16	13	22	17	50	3
	2	14	14	13	19	15	60	1
	3	19	19	20	23	*M*	50	0
	4(*D*)	*M*	0	*M*	[0]	0	50	0
Demand		30	20	70	30	60	Select $x_{44} = 30$	
Column difference		2	14	0	(19)	15	Eliminate column 4	

		Destination					
		1	**2**	**3**	**5**	**Supply**	**Row Difference**
Source	1	16	16	13	17	50	3
	2	14	14	13	15	60	1
	3	19	19	20	*M*	50	0
	4(*D*)	*M*	0	*M*	[0]	20	0
Demand		30	20	70	60	Select $x_{45} = 20$	
Column difference		2	14	0	(15)	Eliminate row 4(*D*)	

Transportation Simplex method

		Destination					
		1	2	3	5	Supply	Row Difference
Source	1	16	16	[13]	17	50	(3)
	2	14	14	13	15	60	1
	3	19	19	20	*M*	50	0
Demand		30	20	70	40	Select $x_{13} = 50$	
Column difference		2	2	0	2	Eliminate row 1	

		Destination					
		1	2	3	5	Supply	Row Difference
Source	2	14	14	13	[15]	60	1
	3	19	19	20	*M*	50	0
Demand		30	20	20	40	Select $x_{25} = 40$	
Column difference		5	5	7	(*M* − 15)	Eliminate column 5	

		Destination				
		1	2	3	Supply	Row Difference
Source	2	14	14	[13]	20	1
	3	19	19	20	50	0
Demand		30	20	20	Select $x_{23} = 20$	
Column difference		5	5	(7)	Eliminate row 2	

Transportation Simplex method

		Destination 1	2	3	Supply	Row Difference
Source	2	14	14	[13]	20	1
	3	19	19	20	50	0
Demand		30	20	20	Select $x_{23} = 20$	
Column difference		5	5	(7)	Eliminate row 2	

		Destination 1	2	3	Supply	
Source	3	19	19	20	50	
Demand		30	20	0	Select $x_{31} = 30$ $x_{32} = 20$ $x_{33} = 0$	$Z = 2{,}460$

3. *Russell's approximation method:* For each source row i remaining under consideration, determine its $\bar{u}_i$, which is the largest unit cost c_{ij} still remaining in that row. For each destination column j remaining under consideration, determine its $\bar{v}_j$, which is the largest unit cost c_{ij} still remaining in that column. For each variable x_{ij} not previously selected in these rows and columns, calculate $\Delta_{ij} = c_{ij} - \bar{u}_i - \bar{v}_j$. Select the variable having the *largest* (in absolute terms) *negative* value of Δ_{ij}.

Initial BF solution from Russell's approximation method

Iteration	$\bar{u}_1$	$\bar{u}_2$	$\bar{u}_3$	$\bar{u}_4$	$\bar{v}_1$	$\bar{v}_2$	$\bar{v}_3$	$\bar{v}_4$	$\bar{v}_5$	Largest Negative Δ_{ij}	Allocation
1	22	19	M	M	M	19	M	23	M	$\Delta_{45} = -2M$	$x_{45} = 50$
2	22	19	M		19	19	20	23	M	$\Delta_{15} = -5 - M$	$x_{15} = 10$
3	22	19	23		19	19	20	23		$\Delta_{13} = -29$	$x_{13} = 40$
4		19	23		19	19	20	23		$\Delta_{23} = -26$	$x_{23} = 30$
5		19	23		19	19		23		$\Delta_{21} = -24$*	$x_{21} = 30$
6										Irrelevant	$x_{31} = 0$
											$x_{32} = 20$
											$x_{34} = 30$
											$Z = 2{,}570$

*Tie with $\Delta_{22} = -24$ broken arbitrarily.

3. *Russell's approximation method:* For each source row i remaining under consideration, determine its $\bar{u}_i$, which is the largest unit cost c_{ij} still remaining in that row. For each destination column j remaining under consideration, determine its $\bar{v}_j$, which is the largest unit cost c_{ij} still remaining in that column. For each variable x_{ij} not previously selected in these rows and columns, calculate $\Delta_{ij} = c_{ij} - \bar{u}_i - \bar{v}_j$. Select the variable having the *largest* (in absolute terms) *negative* value of Δ_{ij}.

Initial transportation simplex tableau (before we obtain $c_{ij} - u_i - v_j$) from Russell's approximation method

Iteration 0		Destination 1	2	3	4	5	Supply	u_i
Source	1	16	16	13 (40)	22	17 (10)	50	
	2	14 (30)	14	13 (30)	19	15	60	
	3	19 (0)	19 (20)	20	23 (30)	M	50	
	4(D)	M	0	M	0	0 (50)	50	
Demand		30	20	70	30	60	$Z = 2{,}570$	
	v_j							

Optimality test: A BF solution is optimal if and only if $c_{ij} - u_i - v_j \geq 0$ for every (i, j) such that x_{ij} is nonbasic

Thus, the only work required by the optimality test is the derivation of the values of u_i and v_j for the current BF solution and then the calculation of these $c_{ij} - u_i - v_j$ as described below.

Since $c_{ij} - u_i - v_j$ is required to be zero if x_{ij} is a basic variable, u_i and v_j satisfy the set of equations $c_{ij} - u_i - v_j = 0$ for each (i, j) such that x_{ij} is basic.

There are $m + n - 1$ basic variables, and so there are $m + n - 1$ of these equations. Since the number of unknowns (the u_i and v_j) is $m + n$, one of these variables can be assigned a value arbitrarily without violating the equations. The choice of this one variable and its value does not affect the value of any $c_{ij} - u_i - v_j$ even when x_{ij} is nonbasic, so the only (minor) difference it makes is in the ease of solving these equations. A convenient choice for this purpose is to select the u_i that has the *largest number of allocations in its row* (break any tie arbitrarily) and to assign to it the value zero. Because of the simple structure of these equations, it is then very simple to solve for the remaining variables algebraically.

To demonstrate, we give each equation that corresponds to a basic variable in our initial BF solution.

x_{31}:	$19 = u_3 + v_1$.	Set $u_3 = 0$, so $v_1 = 19$,
x_{32}:	$19 = u_3 + v_2$.	$v_2 = 19$,
x_{34}:	$23 = u_3 + v_4$.	$v_4 = 23$.
x_{21}:	$14 = u_2 + v_1$.	Know $v_1 = 19$, so $u_2 = -5$.
x_{23}:	$13 = u_2 + v_3$.	Know $u_2 = -5$, so $v_3 = 18$.
x_{13}:	$13 = u_1 + v_3$.	Know $v_3 = 18$, so $u_1 = -5$.
x_{15}:	$17 = u_1 + v_5$.	Know $u_1 = -5$, so $v_5 = 22$.
x_{45}:	$0 = u_4 + v_5$.	Know $v_5 = 22$, so $u_4 = -22$.

Setting u_3 0 (since row 3 of Table in slide 19 has the largest number of allocations—3) and moving down the equations one at a time immediately give the derivation of values for the unknowns shown to the right of the equations. (Note that this derivation of the u_i and v_j values depends on which x_{ij} variables are *basic variables* in the current BF solution, so this derivation will need to be repeated each time a new BF solution is obtained.)

Transportation Simplex method

We are now in a position to apply the optimality test by checking the values of $c_{ij} - u_i - v_j$ given in Table below. Because two of these values ($c_{25} - u_2 - v_5 = -2$ and $c_{44} - u_4 - v_4 = -1$) are negative, we conclude that the current BF solution is not optimal.

Therefore, the transportation simplex method must next go to an iteration to find a better BF solution.

Completed initial transportation simplex tableau

Iteration 0		Destination 1	2	3	4	5	Supply	u_i
Source	1	16 (+2)	16 (+2)	13 (40)	22 (+4)	17 (10)	50	−5
	2	14 (30)	14 (0)	13 (30)	19 (+1)	15 (−2)	60	−5
	3	19 (0)	19 (20)	20 (+2)	23 (30)	M ($M-22$)	50	0
	4(*D*)	M ($M+3$)	0 (+3)	M ($M+4$)	0 (−1)	0 (50)	50	−22
Demand		30	20	70	30	60	$Z = 2{,}570$	
	v_j	19	19	18	23	22		

An Iteration

As with the full-fledged simplex method, an iteration for this streamlined version must determine an entering basic variable (step 1), a leaving basic variable (step 2), and then identify the resulting new BF solution (step 3).

Step 1: Find the Entering Basic Variable. Since $c_{ij} - u_i - v_j$ represents the rate at which the objective function will change as the nonbasic variable x_{ij} is increased, the entering basic variable must have a *negative* $c_{ij} - u_i - v_j$ value to decrease the total cost Z. Thus, the candidates in Table (previous slide) are x_{25} and x_{44}. To choose between the candidates, select the one having the larger (in absolute terms) negative value $c_{ij} - u_i - v_j$ to be the entering basic variable, which is x_{25} in this case.

Step 2: Find the Leaving Basic Variable. Increasing the entering basic variable from zero sets off a *chain reaction* of compensating changes in other basic variables (allocations), in order to continue satisfying the supply and demand constraints. The first basic variable to be decreased to zero then becomes the leaving basic variable.

Part of initial transportation simplex tableau showing the chain reaction caused by increasing the entering basic variable x_{25}

		Destination						Supply
			3		4		5	
Source	1	…	13	40 +	22	+4	17 10 −	50
	2	…	13	30 −	19	+1	15 +	60
		…	…		…		…	
Demand			70		30		60	

Step 3: Find the New BF Solution. The *new BF solution* is identified simply by adding the value of the leaving basic variable (before any change) to the allocation for each recipient cell and subtracting *this same amount* from the allocation for each donor cell. In Table previous slide the value of the leaving basic variable x_{15} is 10, so the portion of the transportation simplex tableau in this table changes as shown in Table below for the new solution. (Since x_{15} is nonbasic in the new solution, its new allocation of zero is no longer shown in this new tableau.)

Part of second transportation simplex tableau showing the changes in the BF solution

		Destination						Supply
			3		4		5	
Source	1	…	13 (50)		22		17	50
	2	…	13 (20)		19		15 (10)	60
		…	…		…		…	
Demand			70		30		60	

Tutorial 7: Transportation Problems

Problem 1

Consider the transportation problem having the following parameter table:

Cost Per Unit Distributed				
		Destination		
		1	2	3	Supply
Source	1	15	9	13	7
	2	11	1M	17	5
	3	9	11	9	3
Demand		7	3	5	

(a) Use Vogel's approximation method manually to select the first basic variable for an initial BF solution.

(b) Use Russell's approximation method manually to select the first basic variable for an initial BF solution.

(c) Use the northwest corner rule manually to construct a complete initial BF solution.

Problem 1

Consider the transportation problem

		Unit Shipping Cost Retail Outlet			
		1	**2**	**3**	**4**
Plant	1	$500	$600	$400	$200
	2	$200	$900	$100	$300
	3	$300	$400	$200	$100
	4	$200	$100	$300	$200

Plants 1, 2, 3, and 4 make 10, 20, 20, and 10 shipments per month, respectively. Retail outlets 1, 2, 3, and 4 need to receive 20, 10, 10, and 20 shipments per month, respectively. The distribution manager, Randy Smith, now wants to determine the best plan for how many shipments to send from each plant to the respective retail outlets each month. Randy's objective is to minimize the total shipping cost.

(a) Formulate this problem as a transportation problem by constructing the appropriate parameter table.

(b) Use the northwest corner rule to construct an initial BF solution.

(c) Starting with the initial basic solution from part (*b*), interactively apply the transportation simplex method to obtain an optimal solution.